AGENTIC AI PROJECT

Project Guidelines and Instructions to Students

1. Purpose of the Project

As part of the Genrative AI **course**, students are required to develop and implement an **Agentic AI-based mini project**.

The objective of this project is to provide hands-on experience in designing intelligent systems that can understand a goal, plan tasks, use tools, interact with other agents, retrieve knowledge, maintain context, and generate an appropriate final response.

The project should demonstrate the practical application of concepts covered in the course.

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3. Project Requirements

Each project must demonstrate the following Agentic AI characteristics:

Mandatory Components

1. Large Language Model (LLM)

Students may use an appropriate LLM such as:

- OpenAI models
- Gemini
- Llama
- Mistral
- Qwen
- Other approved LLMs
- Local LLMs using LM Studio or similar platforms

2. Agent

The system must contain at least one functional AI agent.

3. Planning / Task Decomposition

The agent should be able to divide a user goal into multiple tasks wherever applicable.

4. Tool Usage

At least **two tools, APIs, databases, or external resources** should be integrated.

Examples:

- Web search
- Python
- Calculator
- Database
- PDF/document reader
- External API
- Weather API
- File processing
- Knowledge base

5. Memory / Context

The system should maintain relevant conversation context or other forms of memory.

6. RAG

Where applicable, students should implement Retrieval-Augmented Generation using their own documents or a suitable knowledge base.

7. Multi-Agent Communication

Projects involving multiple agents should demonstrate communication and information exchange between agents.

8. Validation / Error Handling

The system should handle incorrect, incomplete, or unavailable information appropriately.

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4. Suggested Project Areas

Students may select a project from the following areas or propose another suitable topic with faculty approval.

S.No.	Suggested Project
1	AI Student Academic Assistant
2	Research Paper Assistant
3	Placement Preparation Agent
4	Resume Analysis and Career Assistant
5	AI Travel Planning Agent
6	Customer Support Multi-Agent System
7	E-Commerce Shopping Assistant
8	AI Personal Finance Assistant
9	Healthcare Information Assistant

S.No.	Suggested Project
10	Smart Agriculture Agent
11	College Admission Assistant
12	Library Management Agent
13	AI Coding and Debugging Assistant
14	Meeting Assistant
15	Online Learning Assistant
16	Smart City Complaint Management Agent
17	College Event Management Agent
18	News Analysis and Summarization Agent
19	Disaster Management Multi-Agent System
20	Legal Document Information Assistant

5. Important Project Rule

A simple chatbot will NOT be considered an Agentic AI project.

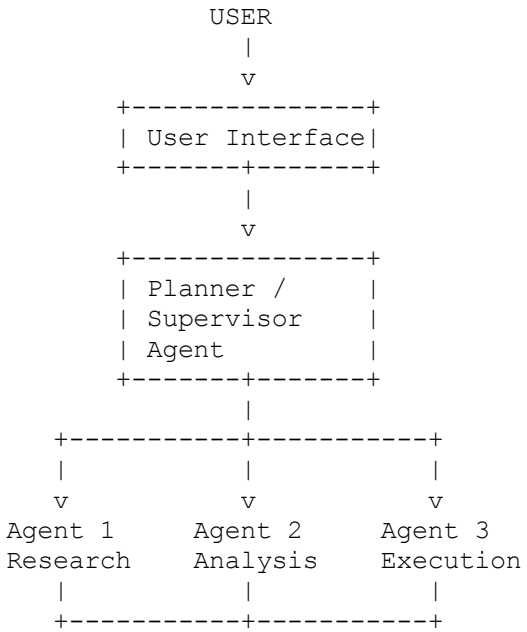
The project should demonstrate an agentic workflow such as:

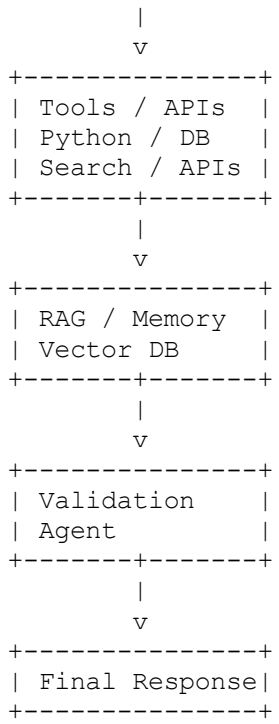
User Goal → Planning → Task Decomposition → Tool Use → Knowledge Retrieval → Agent Execution → Validation → Final Response

A project that only sends a prompt to an LLM and displays the generated response will not satisfy the project requirements.

6. Recommended System Architecture

Students are encouraged to follow an architecture similar to the following:





The actual architecture may vary depending on the selected project.

7. Recommended Technologies

Students may select technologies based on their project requirements.

Programming Language

- Python – Recommended
- JavaScript / Node.js – Optional

Agent Frameworks

Students may use:

- LangGraph
- LangChain
- CrewAI
- AutoGen
- Other approved agent frameworks

LLM

- OpenAI
- Gemini
- Llama
- Mistral
- Qwen

- LM Studio/local LLMs
- Other approved models

Vector Databases

Examples:

- FAISS
- ChromaDB
- Pinecone
- Other suitable vector databases

Interface

- Streamlit
- Gradio
- React
- Web application
- Other suitable interface

Students should clearly mention all technologies and versions used in the project.

8. Project Development Stages

Students should complete the project in the following stages.

Phase 1 – Problem Identification

- Identify a real-world problem.
- Define the problem statement.
- Identify users and requirements.
- Explain why Agentic AI is suitable.

Phase 2 – Literature Survey

Study existing systems, research papers, frameworks, and applications related to the selected topic.

Minimum recommended:

5–10 relevant references

Phase 3 – System Design

Prepare:

- System architecture
- Agent architecture

- Workflow
- Data flow
- Agent communication diagram
- Use case diagram
- Sequence diagram

Phase 4 – Implementation

Implement:

- LLM integration
- Agent(s)
- Planning
- Tool usage
- RAG
- Memory
- Agent communication
- Error handling
- User interface

Phase 5 – Testing

Prepare test cases covering:

- Normal inputs
- Multiple-step tasks
- Tool invocation
- Knowledge retrieval
- Invalid inputs
- Error conditions

Phase 6 – Evaluation

Evaluate the system using appropriate metrics.

Phase 7 – Documentation

Prepare the final project report according to the format given below.

Phase 8 – Demonstration

Demonstrate the working prototype and explain the complete architecture.

9. Project Report Format

The final report should contain the following sections.

Cover Page

Include:

- Project Title
- Agentic AI Course Project
- Student Name(s)
- Roll Number(s)
- Branch / Section
- Academic Year
- Course Name
- Faculty Name
- Institution Name

Chapter 1 – Introduction

1.1 Background

1.2 Problem Statement

1.3 Motivation

1.4 Objectives

1.5 Scope of the Project

Chapter 2 – Literature Survey

Discuss existing research and systems related to the selected project.

Include a comparison table:

Author Year Method Application Advantages Limitations

Students should identify the **gap or limitation in existing approaches** that motivates their proposed system.

Chapter 3 – Problem Definition and Requirements

3.1 Existing System

3.2 Limitations of Existing System

3.3 Proposed System

3.4 Functional Requirements

3.5 Non-Functional Requirements

3.6 Hardware Requirements

3.7 Software Requirements

Chapter 4 – System Design

This chapter should explain the proposed Agentic AI architecture.

Include:

4.1 Overall Architecture

4.2 Agent Architecture

4.3 Agent Roles and Responsibilities

4.4 Task Planning and Decomposition

4.5 Tool Integration

4.6 RAG Architecture

4.7 Memory Architecture

4.8 Agent Communication

4.9 Data Flow Diagram

4.10 Sequence Diagram

4.11 Use Case Diagram

Chapter 5 – Implementation

Explain the actual implementation.

Include:

5.1 LLM Integration

5.2 Prompt Design

5.3 Agent Implementation

5.4 Planning Mechanism

5.5 Tool Integration

5.6 RAG Implementation

5.7 Memory Implementation

5.8 Multi-Agent Communication

5.9 Error Handling

5.10 User Interface

Students should include relevant screenshots and explain each major component.

Chapter 6 – Testing and Results

Students must provide evidence that the system works correctly.

Test Case Format

Test Case	Input	Expected Output	Actual Output	Status
TC01	User query	Correct response	Obtained response	Pass
TC02	Multi-step request	Tasks executed	Tasks executed	Pass
TC03	Document query	Relevant information	Retrieved information	Pass
TC04	Invalid input	Appropriate error	Error handled	Pass

Students should include screenshots of important outputs.

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10. Performance Evaluation

Students should select appropriate metrics for their project.

Possible metrics include:

Metric	Description
Task Success Rate	Percentage of tasks successfully completed
Response Accuracy	Correctness of generated response
Retrieval Accuracy	Quality of retrieved information
Response Time	Time required to complete a task
Tool Success Rate	Successful tool execution
Hallucination Rate	Incorrect/generated unsupported information
User Satisfaction	User evaluation of the system
Token Usage	LLM token consumption
Cost per Task	Approximate API cost

Not every metric is mandatory. Students should select metrics relevant to their project.

Chapter 7 – Conclusion and Future Scope

7.1 Conclusion

Summarize the work completed and objectives achieved.

7.2 Limitations

Clearly mention current limitations.

7.3 Future Scope

Discuss possible improvements such as:

- Voice-based agents
- Multimodal agents
- Additional tools
- Advanced memory
- Human-in-the-loop systems
- Local LLM deployment
- Cloud deployment
- Mobile application
- Additional autonomous capabilities

Chapter 8 – References

Use a consistent citation style.

Students should include:

- Research papers
- Books
- Official documentation
- Dataset sources
- APIs
- Framework documentation

All references cited in the report should also appear in the reference list.

14. Final Submission Requirements

Each group must submit:

1. Project Report

PDF format

2. Source Code

Complete and executable source code. With deployment link